YASSIN ELMONNIAR

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Biomedical Engineering graduate studying an MSc in Artificial Intelligence & Machine Learning at the University of Portsmouth. Practical experience across machine learning, computer vision, NLP, medical imaging, EPR and hospital information systems. Seeking AI/ML and Medical AI roles combining software, data and biomedical engineering for healthcare technologies.

Education

University of Portsmouth, MSc Artificial Intelligence & Machine Learning, Sep 2026 - Sep 2027 (Expected)
MSc work spans machine learning, deep learning, computer vision, NLP and AI deployment, with emphasis on healthcare and medical imaging.

Machine Learning & Deep Learning

- Apply Python, R, NumPy and scikit-learn to EDA, feature engineering and predictive modelling; train CNNs and transfer-learning models with PyTorch, TensorFlow and Keras.

Computer Vision, NLP & Medical AI

- Implement image-classification and NLP solutions with OpenCV, Hugging Face Transformers and spaCy, including NER and LLM applications.

AI Software Development

- Engineer FastAPI/Flask services and Streamlit/React interfaces, combining Docker, Git/GitHub and pytest for repeatable model serving and testing.

Mansoura University, BEng Biomedical Engineering, 2026
Built a foundation in medical devices, imaging, instrumentation, sensors and digital health, with an early focus on medical AI.

Biomedical Engineering, EPR & Digital Health

- Applied knowledge of MRI systems, instrumentation, EPR and hospital information systems across patient data, clinical documentation and digital workflows.
- Integrated Raspberry Pi and NVIDIA Jetson with Python and C++ for sensor monitoring and hardware-software interfacing.

Healthcare AI Portfolio Development (2025-2026)

- **Heart Disease Risk Assessment:** modelled 303 patient records across 13 predictors; logistic regression reached 86.9% test accuracy.
- **Chest X-ray Classification:** tested a NORMAL/PNEUMONIA classifier on 624 images, achieving 83.3% accuracy and 0.946 ROC-AUC.
- **NLP Medical Consultation Chatbot:** combined 2 NLP components - spaCy NER and Hugging Face Transformers - for medical entity extraction and question answering.
- **MedIntel AI:** integrated 3 AI modules behind 5 FastAPI endpoints and a Streamlit interface, supported by 3 automated API tests and Docker.

Graduation Project - MRI Helium Leak Detection Using IoT Technology (A+)

- Designed an IoT monitoring system around 3 functions: continuous MRI-environment sensing, central monitoring and automated alerts for early leak detection.
- Integrated 3 engineering domains - MRI technology, biomedical engineering and software development - into a preventive-maintenance healthcare application.

Relevant Experience

Information Technology Industry Development Agency (ITIDA), Machine Learning Engineer Trainee, Jun-Jul 2025
Machine Learning & Deep Learning

- Benchmarked ML/DL models with Python, NumPy and scikit-learn across 3 modelling stages: feature extraction, model selection and hyperparameter tuning; explored Apache Spark.
- Validated held-out experiments against 5 measures: accuracy, precision, recall, F1 and ROC analysis.

Post-training Projects (August 2025)

- **Diabetes ML Classifier:** processed 768 records with 8 clinical features; SMOTE expanded the dataset to 1,000 samples and Random Forest reached 85.0% test accuracy after tuning 3 classifiers.
- **Obesity Risk Detection:** prepared 20,758 records across 18 columns for multi-class modelling of lifestyle, physical and demographic variables.
- **Sales Forecasting:** engineered 7 lag features and fitted a 100-tree Random Forest Regressor on a 500-day demo series, assessed with RMSE and R-squared.

Huawei Egyptian Academy, Artificial Intelligence Engineer Trainee, May 2025 - June 2025

Computer Vision & Natural Language Processing

- Applied 2 learning paradigms - supervised and unsupervised - across ML case studies to produce data-driven solutions.
- Configured a 3-stage computer-vision classification pipeline covering image preparation, model training and validation.
- Implemented NLP applications across 2 task areas - entity extraction and language processing - and reviewed outputs against requirements.

Parallel MRI Project (May 2025)

- **MRI Visualizer:** built an interactive Python tool for NIfTI and DICOM, supporting 2D/3D exploration across 3 anatomical planes.
- Added 3 image-inspection controls: slice scrolling, contrast adjustment and optional mask overlays, supporting preprocessing.

National Telecommunications Institute (NTI), Machine Learning & Data Science Trainee, May 2023 - July 2023

Data Science & Predictive Modelling

- Trained predictive models with Python and scikit-learn across 3 modelling stages: data preparation, model selection and validation.
- Prepared datasets through 3 preprocessing stages: cleaning, encoding and feature scaling before model training.
- Compared supervised and unsupervised approaches across 3 task types: classification, regression and clustering.
- Analysed datasets with EDA and 2 visualisation libraries - Matplotlib and Seaborn - summarising findings for project review.

Courses

Machine Learning - Stanford University / Andrew Ng, December 2022
- Supervised/unsupervised learning, model evaluation and core ML methods.

Applied Data Science with Python Specialization - University of Michigan, April 2023
- Python data analysis, visualisation and machine-learning methods.

Google Data Analytics Professional Certificate - Google, February 2024
- Data preparation, analysis, visualisation and reporting.

IBM Data Science Professional Certificate - IBM, August 2024
- Python, data-science and machine-learning project work.

Deep Learning Specialization - DeepLearning.AI, August 2024
- Neural networks, CNNs, model tuning and deep-learning methods.

Generative AI with Large Language Models - DeepLearning.AI & AWS, January 2025
- Transformers, LLM applications and generative-AI techniques.

AI for Everyone - DeepLearning.AI / Andrew Ng, May 2025
- AI concepts, business use cases and responsible adoption.

AI and Machine Learning Summer Camp - Microsoft, July 2025
- Practical AI and machine-learning use cases.

Additional Information

- **Programming & Data:** Python, R, SQL, C++, pandas, NumPy and scikit-learn.
- **Deep Learning & Computer Vision:** PyTorch, TensorFlow, Keras, OpenCV, CNNs and transfer learning.
- **NLP & Generative AI:** Hugging Face Transformers, spaCy, NER, transformer models and LLM applications.
- **AI Software & Deployment:** FastAPI, Flask, REST APIs, Streamlit, React, Docker, Git/GitHub, pytest and Apache Spark.
- **Biomedical & Digital Health:** MRI systems, medical imaging, EPR, hospital information systems, Raspberry Pi, NVIDIA Jetson and PyQt.
- **Engineering & Analytics:** Excel, Power BI, Tableau, Matplotlib, AutoCAD, SolidWorks and Fusion 360.
- **Languages:** Arabic (Native); English (Working Proficiency).
- **Mobility:** Based in Portsmouth; open to relocation across the UK and willing to travel as required.

Interests

- Strength training and fitness, with regular gym sessions.
- AI and technology talks, workshops and professional events.
- Technical presentations, knowledge-sharing and discussion.